

GenAI to explain CNN

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Introduction

The interpretability of convolutional neural networks (CNNs) represents one of the most current and relevant challenges in the field of deep learning. This work proposes an approach that combines generative AI techniques and semantic analysis tools to explore and explain the internal functioning of a CNN.

In particular, we use the generative model **Stable Diffusion 3.5 Large** to create a synthetic image dataset based on lemmas selected through **Core WordNet**. These images are then used as input for the **ResNet-50** network, whose internal activations are analyzed through the use of *forward hooks*. This allows us to relate activation patterns to well-defined semantic concepts represented by WordNet synsets.

As a complement to this first analysis, the true challenge of this work lies in demonstrating and evaluating the presence of bias in the predictions produced by the CNN. To achieve this, five particularly representative classes were selected—those known to be especially prone to bias, as reported in **Salient ImageNet**. We then generated structured prompts in three distinct modes: “object-only,” “context-only,” and “mixed.”

The results show that the deeper layers of the CNN are able to effectively discriminate between the proposed semantic concepts, confirming the model’s ability to learn complex representations. The bias analysis, on the other hand, reveals a significant dependence of the network on the visual context: the presence of contextual elements in the prompts significantly increases the confidence of the predictions.

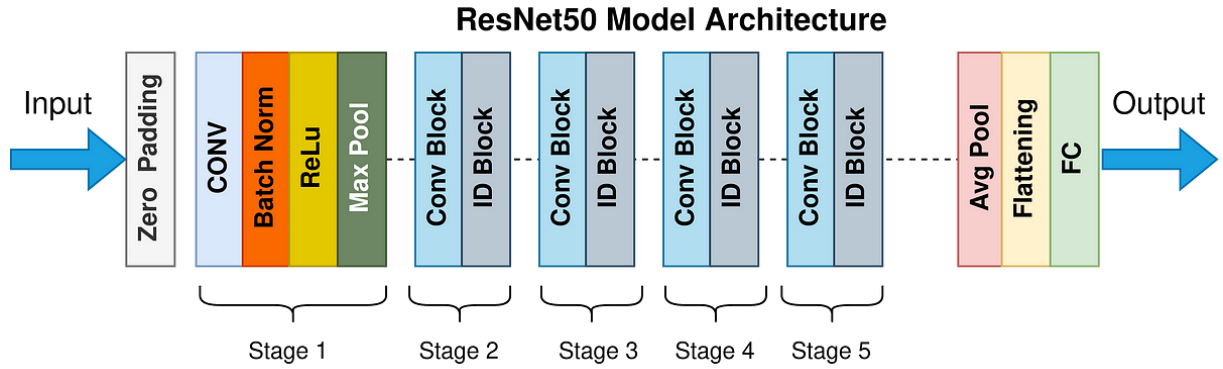


Figure 1: Architecture of *ResNet-50*

1. Objectives and Methodology

1.1 Interpretative Analysis of the CNN

Before addressing the issue of bias, the goal was to deeply explore the internal functioning of the convolutional neural network ResNet-50, with the aim of understanding how, and to what extent, its layers represent and distinguish semantic concepts. To this end, a methodology was adopted involving the generation of a synthetic image dataset, built from lemmas selected through Core WordNet and described using generic, pre-defined textual prompts. These images were then used as input for ResNet-50, where the internal activations of the main layers were recorded through the use of forward hooks. To quantify these aspects, the following metrics were employed:

- **Maximum activation per synset and layer:** for each layer of the network, the maximum activation value associated with each semantic concept was computed, thus identifying the neurons most sensitive to specific synsets (i.e., what they have specialized in).
- **Neuron-synset correlation:** for the most active neurons in each layer, activation statistics (mean, standard deviation, number of samples) were analyzed with respect to the different synsets, highlighting potential specializations.
- **Semantic separability:** the average distance between the centroids of synset activations across layers was measured, providing a quantitative index of the layer’s ability to distinguish between different concepts.

1.2 Bias Analysis

Focusing now on the problem of bias within CNNs, the main goal was to assess the model’s ability to generalize across recognition classes — in particular, to determine the extent to which the predictions of the model, in our case **ResNet-50**, depend on contextual elements rather than on the actual object of interest. Based on the data from **Salient ImageNet**, five classes were selected — **bee**, **frog**, **plane**, **guitar**, and **strawberry** — known to be frequently associated with recognizable environments (flowers for bees, ponds for frogs, sky for airplanes, stages for guitars, plants for strawberries).

For clarity, examples of spurious features for these bias-prone classes are shown below — that is, contextual elements that the CNN “looks at” to aid classification, but which are not intrinsic features of the object itself, merely background cues:



Figure 2: Example of spurious features for *bee*



Figura 3: Example of spurious features for *guitar*



Figura 4: Example of spurious features for *strawberry*

1.3 Prompt Design

For each class, 8 prompts were generated (listed in Table 1) and divided into three categories:

- **Object-only:** descriptions focused solely on the object (e.g., “A professional close-up photograph of a bee with detailed anatomical features.”).
- **Context-only (spurious):** descriptions of the typical context of the object, without mentioning the object itself (e.g., “A vibrant garden full of colorful flowers.”).
- **Mixed:** combinations of object and context to reproduce realistic scenes (e.g., “A bee in a beautiful garden setting with colorful flowers.”).

This approach allows us to evaluate when, and if, the network “cheats” by relying on unnecessary contextual cues that are often strongly correlated with the object.

It is important to highlight a key aspect related to this: **ResNet-50**, like other CNNs, was trained on **ImageNet**, and the spurious features on which it bases its prediction confidence can be traced back to the subject–context correlations it inadvertently learned during training on this dataset. Therefore, our goal here is to determine how far we can push the model’s generalization ability, and whether it truly uses context to boost its predictions.

Tabella 1: Prompts used for each bias class and prompt type.

Class	Object-only	Context-only	Mixed
bee	– A professional close-up photograph of a bee with detailed anatomical features and natural coloring – A bee on a clean white background showing intricate wing and body structure – A simple illustration of a bee isolated against a plain background	– A vibrant garden full of colorful flowers and greenery – A close-up of pollen-covered blossoms in a sunlit garden – A field of daisies and sunflowers under bright daylight	– A bee in a beautiful garden setting with colorful flowers and natural lighting – A bee collecting nectar from blooming flowers in a sunny meadow
frog	– A professional wildlife photograph of a frog with detailed skin texture and natural markings – A frog on a clean white background showing intricate skin patterns – A simple illustration of a frog isolated against a plain background	– A tranquil pond scene with aquatic plants and lily pads – A marshland with crystal clear water and lush vegetation – A wetland ecosystem with reeds and water lilies	– A frog sitting near a crystal clear pond with lush marsh vegetation – A frog resting on lily pads in a serene wetland environment
plane	– A professional close-up photograph of a plane with detailed engineering design – A plane on a clean white background showing aerodynamic features – A simple illustration of a plane isolated against a plain background	– A clear blue sky filled with puffy white clouds – An airport runway with control tower in the distance – A vast blue sky with scattered cumulus clouds at sunset	– A plane flying through clear blue sky with puffy white clouds in perfect weather conditions – A plane approaching the runway at a busy international airport

Class	Object-only	Context-only	Mixed
guitar	– A professional product photograph of a guitar with detailed craftsmanship and wood grain texture – A guitar on a clean white background showing its shape and strings – A simple illustration of a guitar isolated against a plain background	– A concert stage with dramatic stage lights and microphone setup – A music venue with amplifiers and performance equipment – A recording studio with professional audio equipment and lights	– A guitar on a concert stage with dramatic stage lights and microphone setup for performance – A guitar in a recording studio with professional lighting and audio equipment
strawberry	– A professional macro photograph of a strawberry with detailed surface texture and natural red coloring – A strawberry on a clean white background showing its seeds and surface – A simple illustration of a strawberry isolated against a plain background	– A farm field full of lush green plants and ripe berries – An agricultural setting with rows of leafy plants in rich soil – A garden bed with green foliage and growing vegetation	– A strawberry growing on a healthy green plant with leaves in a farm field setting – A strawberry ripening on its plant surrounded by lush green foliage

Here are some examples of the three types of images generated based on the adopted prompts:



Figura 5: Example of *object-only* image



Figura 6: Example of *context-only* image



Figura 7: Example of *mixed* image

1.4 Operational Phases

The following section details all the operational phases characterizing the analysis, structured in two main modes: CNN interpretability on generic classes and bias analysis on selected classes. Although the purposes differ, both aspects are fully integrable, as demonstrated by the obtained results. For greater clarity, refer to Figure 8.

1. **Class selection and prompt generation:** Initially, the classes of interest were selected. For interpretative analysis, generic synsets from Core WordNet were used, while for bias analysis, five bias-aware classes from Salient ImageNet were adopted (**bee**, **frog**, **plane**, **guitar**, **strawberry**). For each class, textual prompts were generated: in the interpretative mode, descriptive and neutral prompts were used, whereas in the bias-aware mode, structured prompts in three categories (object-only, context-only, mixed) were created to isolate contextual effects.
2. **Image generation:** All prompts were provided to Stable Diffusion 3.5 Large, which produced images consistent with the supplied textual descriptions.
3. **Activation analysis:** The generated images were processed by the ResNet-50 network. Using forward hooks (to capture layer outputs), the internal activations of the main CNN layers were extracted, enabling an analysis of how the network represents different semantic concepts. This process applies to both interpretability and bias-aware analyses.
4. **Extraction of classification probabilities:** For each image, the probability assigned by ResNet-50 to the target class was recorded. This allowed a quantitative evaluation of the model’s ability to recognize both generic and bias-aware classes depending on the type of prompt.
5. **Quantitative analysis:** The obtained activations and probabilities were aggregated and compared. For the interpretative mode, metrics such as maximum activation, semantic separability, and layer specialization were assessed. For the bias-aware mode, probabilities for the three prompt categories (object-only, context-only, mixed) were compared, quantifying the impact of context on network prediction and highlighting possible semantic biases for each class.

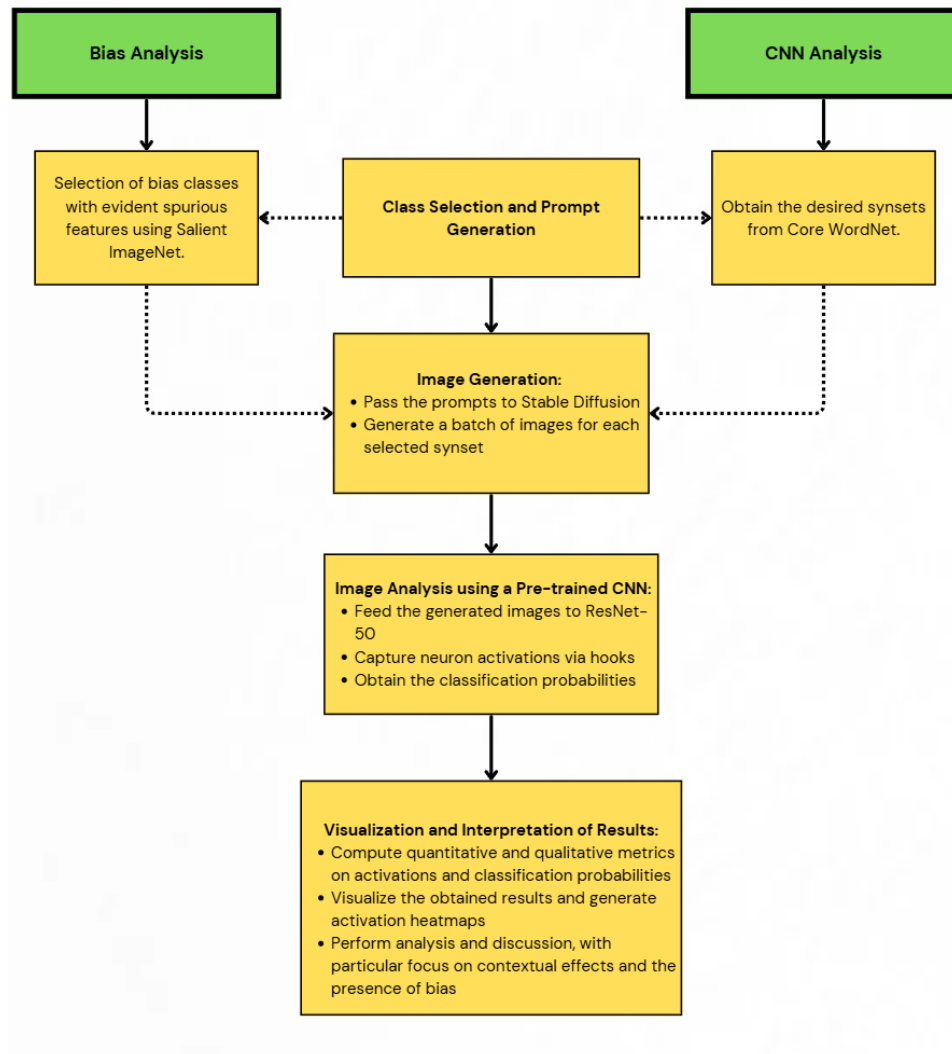


Figura 8: Workflow of the operational phases

2. Code Architecture

The project is structured into five main classes, each with well-defined responsibilities:

1. **CoreWordNetManager**: handles downloading and parsing of the Core WordNet file, selection of synsets (distinguishing between bias-aware and non-bias-aware), generation of both generic and bias-oriented prompts, and management of the WordNet lexicon.
2. **StableDiffusionGenerator**: manages loading of the Stable Diffusion 3.5 Large model and generation of image batches from prompts.
3. **CNNActivationAnalyzer**: loads the ResNet-50 network, preprocesses images, extracts average activations per layer, computes relevant analysis metrics, and visualizes activation patterns.
4. **CNNInterpretabilityProject**: coordinates the entire interpretability pipeline, linking significant activations to semantic concepts extracted from WordNet and producing final metrics and visualizations.
5. **ImageNetBiasAnalyzer**: manages bias classes, performs mapping between ImageNet indices and labels, identifies prompts with the highest predicted probabilities for each bias class, and generates specific bias analysis metrics.

3. Class Functionalities

3.1 Configuration and Parameters (ProjectConfig)

This class centralizes all configuration parameters: directory paths, model names, image dimensions, generation parameters, and the bias classes to be analyzed.

3.2 CoreWordNetManager

- **Download**: the `download_and_parse_from_url()` function downloads and reads the Core WordNet file, extracting a set of lemmas.
- **Synset loading**: `load_synsets_from_wordnet()` loads specific synsets (e.g., `dog.n.01`), excluding bias classes.
- **Fallback**: if insufficient synsets are found, a list of common words is used (also useful for debugging).
- **CNN-pure prompts**: generates descriptive prompts using predefined templates for each semantic category.
- **Bias-aware prompts**: generates prompt groups for each bias class (`bee`, `frog`, `plane`, `guitar`, `strawberry`), divided into object-only, context-only, and mixed.

3.3 StableDiffusionGenerator

- **Loading**: `load_model()` first attempts to load the model in 4-bit quantized mode, falling back to the standard model if needed.
- **Generation**: `generate_image()` and `generate_batch()` generate images from prompts, save the results, and optimize GPU memory management.

3.4 CNNActivationAnalyzer

- **Hooks:** registers forward hooks on the main layers of ResNet-50; these hooks capture activations during the forward pass.
- **Analysis:** computes and prints several key interpretability metrics, including:
 - Maximum activation per synset and layer: reports, for each layer and synset, the maximum activation value and the number of images considered.
 - Neuron/synset correlations: for each layer, shows activation statistics (mean, standard deviation, number of samples) of the most active neurons across synsets.
 - Semantic separability: computes the average distance between synset activation centroids across layers, useful for assessing a layer’s ability to distinguish between different concepts.

3.5 CNNInterpretabilityProject

This class represents the core of the project and integrates all major functionalities:

- Management of the prompt–synset mapping
- Analysis of metrics produced via ResNet hooks
- Centralized execution of experiments, enhancing reproducibility
- Visualization and storage of results

3.6 ImageNetBiasAnalyzer

The `ImageNetBiasAnalyzer` class is responsible for:

- Selecting and managing bias classes
- Performing classification of generated images for prompt groups (object-only, context-only, mixed)
- Identifying and ranking prompts based on the CNN’s predicted probability for each category
- Producing and saving metrics and reports specific to bias analysis

4. Results Obtained

We now analyze the results from an execution that integrates both analyses.

Class Separability in Deep Layers

The heatmap in Figure 9 shows the maximum activation values for each synset across the top 8 layers with the highest semantic separability — that is, the ability to distinguish between the centroids of activations corresponding to different synset classes.

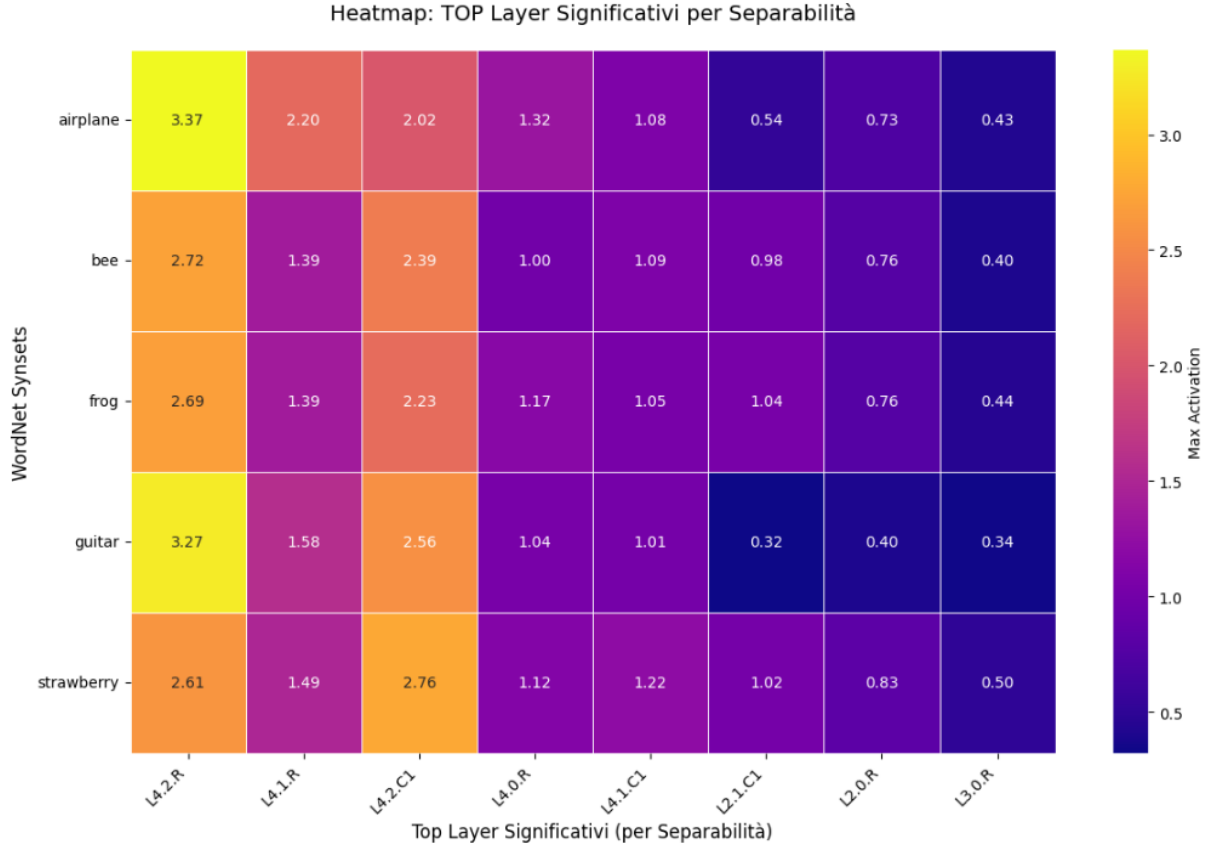


Figura 9: Heatmap for bias classes on the most separable layers

We observe:

- guitar reaches 3.27 on layer4.2.relu.
- airplane reaches 3.37 on layer4.2.relu, followed by 2.20 and 2.02 in previous layers.
- strawberry reaches 2.76 on layer4.2.conv1 and 2.61 on layer4.2.relu.
- bee and frog reach 2.72 and 2.69 in the top layers, but decrease rapidly thereafter.

Quantitative Results

Analyzing the top six layers, we observe that the average separability between classes—computed as the mean distance between activation centroids of different synsets—varies depending on layer depth: in deeper layers, high values are reached (e.g., 16.828 in **layer4.2.relu**), demonstrating greater discriminative ability among the five tested synsets. It is important to note that the global average separability (i.e., across all synset centroids) is always higher than the maximum activations observed for individual synsets, since the global metric considers all possible class pairs. For instance, the global mean distance in **layer4.2.relu** (16.828) is significantly higher than the maximum activation for synsets such as **airplane** or **guitar** in the same layer (3.37 and 3.27, respectively).

Table 1: Average separability per layer (Top 6)

Layer	Avg Centroid Distance
layer4.2.relu	16.828
layer4.1.relu	7.331
layer4.2.conv1	4.443
layer4.0.relu	4.329
layer4.1.conv1	2.952
layer2.1.conv1	2.270

These values are consistent with those shown previously in Figure 9.

Table 2: Top 5 synsets for the 6 most separable layers

Layer	Synset	Max Activation
layer4.2.relu	airplane	3.3669
	guitar	3.2708
	bee	2.7210
	frog	2.6922
	strawberry	2.6121
layer4.1.relu	airplane	2.2011
	guitar	1.5823
	strawberry	1.4935
	bee	1.3906
	frog	1.3889
layer4.2.conv1	strawberry	2.7643
	guitar	2.5606
	bee	2.3852
	frog	2.2322
	airplane	2.0180
layer4.0.relu	airplane	1.3186
	frog	1.1688
	strawberry	1.1203
	guitar	1.0430
	bee	0.9994
layer4.1.conv1	strawberry	1.2223
	bee	1.0875
	airplane	1.0846
	frog	1.0466
	guitar	1.0144
layer2.1.conv1	frog	1.0400
	strawberry	1.0224
	bee	0.9789
	airplane	0.5362
	guitar	0.3222

Comparison with a Previous Run (100 prompts)

A previous run was performed on a different semantic subset, with 10 images for each of 10 selected synsets. The detailed results below show, for some layers, the top 3 synsets with the highest mean activation (i.e., where the layer is most specialized):

- **layer4.1.relu:**
 - tree.n.01: 2.5172
 - boat.n.01: 2.3055
 - flower.n.01: 2.1233
- **layer4.0.relu:**
 - flower.n.01: 1.6804
 - boat.n.01: 1.6279
 - computer.n.01: 1.3162
- **layer3.3.relu:**
 - computer.n.01: 0.4060
 - dog.n.01: 0.3881
 - flower.n.01: 0.3645
- **layer2.2.bn2:**
 - computer.n.01: 0.3557
 - car.n.01: 0.3469
 - cat.n.01: 0.3369
- **layer2.1.bn3:**
 - tree.n.01: 0.2985
 - flower.n.01: 0.2932
 - bicycle.n.01: 0.2736
- **layer1.0.bn2:**
 - computer.n.01: 0.5834
 - dog.n.01: 0.5657
 - chair.n.01: 0.5599
- **layer4.0.bn1:**
 - dog.n.01: 0.1840
 - cat.n.01: 0.1813
 - flower.n.01: 0.1747
- **layer4.2.bn1:**
 - bicycle.n.01: 0.2196
 - tree.n.01: 0.2030
 - chair.n.01: 0.1647
- **layer3.5.conv3:**
 - dog.n.01: 0.1595
 - bird.n.01: 0.1476
 - cat.n.01: 0.1155
- **layer4.2.conv2:**
 - chair.n.01: 0.0890
 - cat.n.01: 0.0845
 - dog.n.01: 0.0830

Observations: This run confirms the same trend: deeper layers (**layer4.***) show higher semantic separability values, highlighting a greater ability to discriminate among concepts even with varied prompts. The most separable classes differ slightly but include diverse concepts such as *dog*, *tree*, and *flower*.

Neuron Activation Examples

To further investigate the internal behavior of the network, we analyzed the mean activations of neurons in response to the generated prompts. Below are three representative examples from `layer4.2.relu`—the most high-level and discriminative layer—chosen for their ability to highlight distinctive patterns across different synsets. Alongside the means, we report standard deviations, which indicate variability across the 8 images for each class.

Table 3: Mean activations and standard deviations for neurons in `layer4.2.relu`

Neuron	Synset	Mean	Std
624	bee	1.569	0.683
	frog	2.616	0.868
	airplane	1.152	1.171
	guitar	2.276	0.710
	strawberry	1.369	0.690
589	bee	1.194	0.607
	frog	1.028	0.606
	airplane	1.487	1.465
	guitar	3.220	1.660
	strawberry	0.518	0.258
617	bee	2.721	1.374
	frog	1.629	1.274
	airplane	0.684	0.278
	guitar	0.487	0.328
	strawberry	1.732	1.211

In these examples, each neuron responds differently to the tested classes. For instance, neuron 624 is particularly active for `frog`, while neuron 617 shows high variability for `bee`. These patterns indicate the presence of neurons specialized in detecting specific visual features characteristic of different synsets.

5. Bias Analysis

The bias-aware tests reveal a clear dependence of the model on the explicit presence of a visual context within the image. In particular, the addition of recognizable and central contextual elements systematically leads to an increase in classification confidence. The analysis of the generated prompts shows the following behavior:

- **Object-only** prompts, in which the target object is presented without any context, generate correct predictions with good confidence. The values range from around 0.61 (**airplane**) to over 0.99 (**strawberry**), confirming that the model is capable of recognizing the class even in the absence of environmental cues. Nevertheless, the opposite can also be observed: for classes such as **bee**, **airplane**, and **strawberry**, some prompts were not correctly classified.
- **Context-only** prompts, which completely remove the object leaving only semantically coherent environments, are generally ignored: confidence is always below 1%, even in strongly associative cases (such as ponds for **frog** or runways for **airplane**). This suggests that the model is not significantly biased toward contextual cues alone.
- **Mixed** prompts, which include both the object and a plausible context, show more variable behavior. For classes such as **bee** and **strawberry**, the context strengthens the prediction, resulting in near-100% confidence. Conversely, for classes such as **guitar**, the addition of context can sometimes reduce confidence, likely because the surrounding environment may be misleading for classification.

The following tables report, for each target class, the confidence values obtained by the model on the different prompts, divided by category:

Tabella 2: Class: **bee**

Prompt	Confidence
<i>Mixed</i>	
A bee in a beautiful garden setting with colorful flowers and lush greenery.	0.9950
A bee collecting nectar from blooming flowers in a sunny meadow.	0.9805
<i>Object-only</i>	
A bee on a clean white background showing intricate wing and body detail.	0.9735
A professional close-up photograph of a bee with detailed anatomy.	0.1645
A simple illustration of a bee isolated against a plain background.	0.0097
<i>Context-only</i>	
A close-up of pollen-covered blossoms in a sunlit garden.	0.2678
A field of daisies and sunflowers under bright daylight.	0.0047
A vibrant garden full of colorful flowers and greenery.	0.0044

Tabella 3: Class: **frog**

Prompt	Confidence
<i>Mixed</i>	
A frog resting on lily pads in a serene wetland environment.	0.9020
A frog sitting near a crystal clear pond with lush marsh vegetation.	0.0698
<i>Object-only</i>	
A frog on a clean white background showing intricate skin patterns.	0.9348
A simple illustration of a frog isolated against a plain background.	0.6945
A professional wildlife photograph of a frog with detailed skin texture.	0.5284
<i>Context-only</i>	
A garden bed with green foliage and growing vegetation.	0.0005
A close-up of pollen-covered blossoms in a sunlit garden.	0.0005
A tranquil pond scene with aquatic plants and lily pads.	0.0004

Tabella 4: Class: **airplane**

Prompt	Confidence
<i>Mixed</i>	
An airplane flying through a clear blue sky with puffy white clouds.	0.7531
An airplane approaching the runway at a busy international airport.	0.9531
<i>Object-only</i>	
An airplane on a clean white background showing aerodynamic features.	0.6493
A professional close-up photograph of an airplane with detailed fuselage.	0.4126
A simple illustration of an airplane isolated against a plain background.	0.7717
<i>Context-only</i>	
An airport runway with control tower in the distance.	0.0555
A clear blue sky filled with puffy white clouds.	0.0005
A vast blue sky with scattered cumulus clouds at sunset.	0.0001

Tabella 5: Class: **guitar**

Prompt	Confidence
<i>Mixed</i>	
A guitar in a recording studio with professional lighting and equipment.	0.9498
A guitar on a concert stage with dramatic stage lights and microphones.	0.0040
<i>Object-only</i>	
A simple illustration of a guitar isolated against a plain background.	0.9789
A guitar on a clean white background showing its shape and strings.	0.9227
A professional product photograph of a guitar with detailed wood grain.	0.0004
<i>Context-only</i>	
A music venue with amplifiers and performance equipment.	0.0090
A recording studio with professional audio equipment and lighting.	0.0030
A concert stage with dramatic stage lights and microphone setup.	0.0143

Tabella 6: Class: **strawberry**

Prompt	Confidence
<i>Mixed</i>	
A strawberry ripening on its plant surrounded by lush green leaves.	0.9998
A strawberry growing on a healthy green plant with leaves in a garden.	0.9979
<i>Object-only</i>	
A strawberry on a clean white background showing its seeds and texture.	0.9988
A professional macro photograph of a strawberry with detailed surface.	0.9856
A simple illustration of a strawberry isolated against a plain background.	0.3255
<i>Context-only</i>	
A close-up of pollen-covered blossoms in a sunlit garden.	0.0072
A garden bed with green foliage and growing vegetation.	0.0007
A tranquil pond scene with aquatic plants and lily pads.	0.0006

Finally, it can be observed that within the ranking of top prompts for each class, context-only prompts sometimes correspond to those originally designed for other classes; nevertheless, they may activate the target class more strongly than those explicitly designed for it. This phenomenon confirms that the CNN tends to base classification also (and in some cases primarily) on the context in which the target class was seen during training.

5.1 Class-wise Analysis

We now examine, for each class, what can be inferred from the results presented above, referring to the tables shown previously.

Bee: The generated images indicate that the bee class is clearly affected by a strong bias: the mixed prompts reach almost 100% confidence, although one specific object-only prompt was correctly classified:



Figura 10: A bee on a clean white background showing intricate wing and body detail.

The other two object-only prompts, however, were not recognized:



Figura 11: A professional close-up photograph of a bee with detailed anatomy.



Figura 12: A simple illustration of a bee isolated against a plain background.

The bias for this class becomes evident when comparing the confidence of these two cases above with the following context-only image, whose confidence is approximately 27%. What drives the network to recognize the bee are precisely the flowers in the foreground, as wider garden scenes yield near-zero confidence values.



Figura 13: A close-up of pollen-covered blossoms in a sunlit garden.

Frog: Similar to **bee**, the **frog** class gains a confidence boost when lily pads are clearly visible in the foreground. When these are absent and the frog is surrounded by a generic pond scene, confidence drops to about 6%, clearly showing bias.



Figura 14: A frog resting on lily pads in a serene wetland environment.

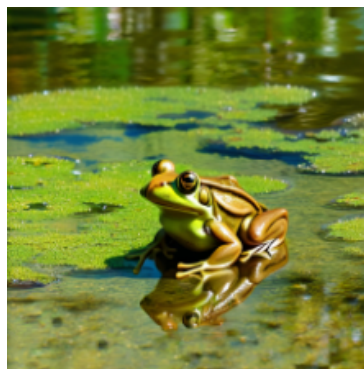


Figura 15: A frog sitting near a crystal clear pond with lush marsh vegetation.

However, unlike **bee**, the object-only prompts achieved excellent results, showing that when the subject is isolated, confidence remains acceptable — suggesting that the context, apart from lily pads, tends to confuse the model rather than help it.

Airplane: The airplane class appears extremely balanced, with high average confidence values in both mixed and object-only cases. A noteworthy exception is a close-up shot of the airplane’s front section, which achieved only about 41% confidence — likely because the network requires more visual cues to confidently identify it as an airplane.



Figure 16: A professional close-up photograph of an airplane with detailed fuselage.

Guitar: It is important to clarify that the ImageNet label analyzed here refers to *electric guitars*, a detail that becomes relevant for the following discussion. Guitars are often associated with concert stages, microphones, and spotlights. Once again, the network’s confidence increases for mixed and context-only prompts, although two particular cases stand out. The first is the following mixed prompt, which produced near-zero confidence — likely because the model becomes uncertain as to which visual features to focus on.



Figure 17: A guitar on a concert stage with dramatic stage lights and microphones.

The second is this object-only image, which also produced a near-zero confidence score. This is probably because it depicts an acoustic, not an electric guitar (our class of interest). Thus, this result is in fact positive, showing that the network does not confuse the two categories even when the context is removed.



Figura 18: A professional product photograph of a guitar with detailed wood grain.

Strawberry: Strawberries are more likely to be recognized when shown on plants or in agricultural environments. Classification was nearly perfect, with the exception of the prompt below, indicating that this class exhibits very low bias.



Figura 19: A simple illustration of a strawberry isolated against a plain background.

6. Conclusions

The internal activation analysis correctly showed that the deepest layers of the network are able to specialize in distinguishing between highly diverse semantic concepts, confirming the model’s ability to learn complex and structured representations.

However, the targeted investigation of biased classes revealed a significant dependence of the model on visual context: the presence of typical environmental elements, often learned during ImageNet training, can substantially increase classification confidence, even in the absence of the target object. Therefore, the research conducted can be considered successful in achieving its goals.

These results highlight that, while CNNs demonstrate strong generalization abilities, they remain vulnerable to spurious correlations and semantic biases introduced by training datasets. The methodology developed in this work has proven effective not only for explaining the internal behavior of the network but also for identifying and quantifying such biases, thus offering a valuable tool for both interpretability research and the development of more robust and reliable models.

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